"allowed" band, then a is not the limit of our sequence.

**Reader:** And when do we reach the certainty?

**Author:** We shall talk this matter over at a later stage because it has nothing to do with the essence of the definition of the limit of the sequence. I suggest that you formulate this definition anew. Don't try to reconstruct the wording given earlier just try to put it in your words. → Imp

**Reader:** I will try. The number a is the limit of a given sequence if for any positive $\varepsilon$ there is (one can find) a serial number n such that for all subsequent numbers (i.e. for the whole "infinite tail" of the sequence) the following inequality holds: $\boxed{|y_n - a| < \varepsilon}$

**Author:** Excellent. You have almost repeated word by word the definition that seemed to you impossible to remember.

**Reader:** Yes, in reality it has turned out to be quite logical and rather easy.

**Author:** It is worthwhile to note that the dialectics of thinking was clearly at work in this case: a concept becomes "not difficult" because the "complexities" built into it were clarified. First, we break up the concept into fragments, expose the "complexities", then examine the "delicate" points, thus trying to reach the "core" of the problem. Then we recompose the concept to make it integral, and as a result, this reintegrated concept becomes sufficiently simple and comprehensible. In the future we shall try first to find the internal structure and internal logic of the concepts and theorems.

I believe we can consider the concept of the limit of sequence as thoroughly analyzed. I should like to add that, as a result, the meaning of the sentence "the sequence converges to a" has been explained. I remind you that initially this sentence seemed to you as requiring no additional explanation.

**Reader:** At the moment it does not seem so self-evident any more. True, I see now quite clearly the idea behind it.

**Author:** Let us get back to examples (5), (7) and (9). These are the sequences that we discussed at the beginning of our talk. To begin with, we note that the fact that a sequence $(y_n)$ converges to a certain number a is conventionally written as $\boxed{\lim_{n\to\infty} y_n = a}$.